Reg. No. | | | | | | | | | | |

[14EC13/23]

FIRST SEMESTER B.TECH EXAMINATION DECEMBER – 2015

BASIC ELECTRONCIS

Time: 3 Hours **Maximum Marks:70**

Instructions : Answer **FIVE** full questions, choosing at least **TWO** from each part.

PART-A

1. a. Explain forward and reverse bias conditions for a P-N junction diode. **07**

   b. Explain Intrinsic and Extrinsic materials with relevant figures. **07**

2. a. In a full wave rectifier. The input is from 30-0-30v transformer. The load and diode forward resistance are 100Ω and 10 Ω respectively. Calculate the average voltage, Rectification efficiency and percentage regulation. **07**

   b. Explain Biased shunt clippers with relevant circuit diagram and waveforms. **04**

   c. For the circuit shown below in fig(a) find the output voltage for the given input shown in figure(b), assuming ideal diode. **03**

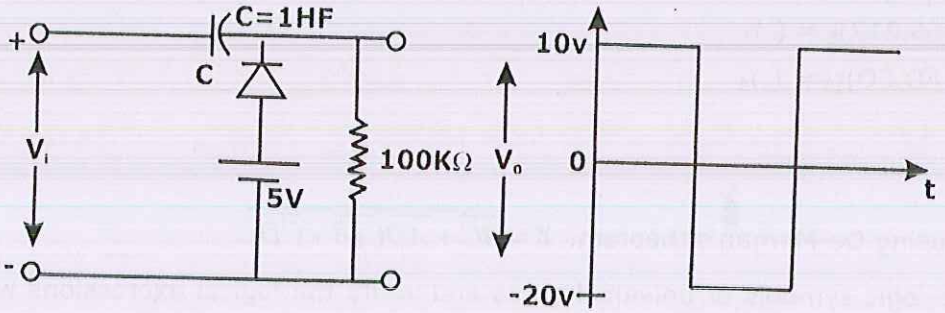

Fig. (a) Fig. (b)

3. a. In a common emitter transistor circuit if β=100, I_B=50MA. Compute the values of α, I_E and I_C. **05**

   b. Draw the common emitter circuit and sketch the i/p and O/P characteristics. Also explain active region, cut-off region and saturation region by indicating them on the characteristic curve. **09**

4. a. Describe the operation of CRO with necessary block diagram. **07**

   b. Explain in brief the characteristics of MOSFET. **07**

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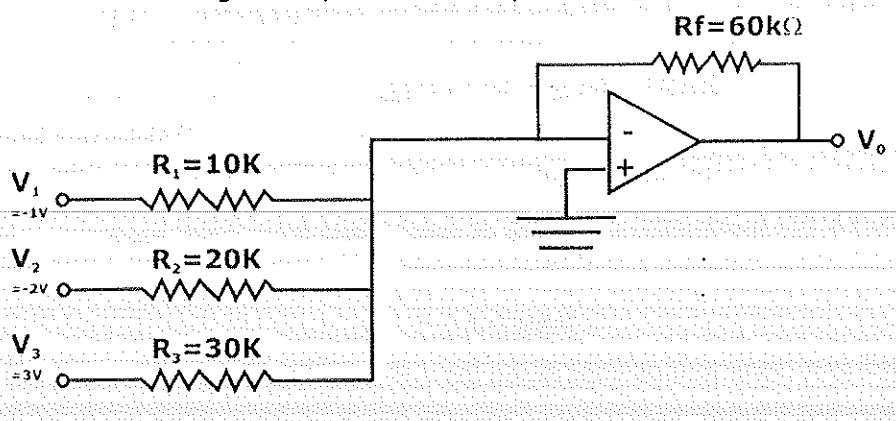
PART-B

5. a. A radio transmitter transmits 10KW of power when percentage modulation is 60. What is carrier power and side band power? **04**

   b. Explain the difference between AM and FM. **05**

   c. Define Modulation. Explain the need for modulation. **05**

6. a. For the following circult, calculate output voltage. **05**



b. Explain the properties of an Ideal Op-Amp. **04**

c. Show how Op-Amp can be used as an Inverting Amplifier. **05**

7. a. i. Subtract $(1111.101)_2$ from $(1001.101)_2$ using 1's complement method. **06**

ii. Subtract $(11101.111)_2$ from $(11111.101)_2$ using 2's complement method.

b. i. $(FACE)_{16} = (\ )_{10}$ **08**

ii. $(29.3749)_{10} = (\ )_2$

iii. $(365.217)_8 = (\ )_2$

iv. $(1AD.EO)_{16} = (\ )_8$

8. a. Simplify using De-Morgan's theorem. $Y = \overline{\overline{BC} + \overline{AD}(\overline{AB} + \overline{CD})}$ **06**

b. Draw the logic symbols of universal gates and verify the logical expressions with truth table. **04**

c. Simplify and implement the following Boolean expression using logic gates. **04**
Y=C(B+C)(A+B+C).